

Summary Video
On Numerical Interpolation

YouTube @StartPracticing

Interpolation : Interpolation is the process of estimating unknown values within the range of known data points

x	y
2	5
4	8
6	13
8	22
10	35

Operators in Interpolation

- 1) Forward difference , $\Delta f(x) = f(x+h) - f(x)$
- 2) Backward difference , $\nabla f(x) = f(x) - f(x-h)$
- 3) Central difference , $\delta f(x) = f(x+\frac{h}{2}) - f(x-\frac{h}{2})$
- 4) Forward shift , $E f(x) = f(x+h)$
- 5) Backward shift , $E^{-1} f(x) = f(x-h)$
- 6) Averaging operator , $\mu f(x) = \frac{1}{2} \left[f(x+\frac{h}{2}) + f(x-\frac{h}{2}) \right]$

Methods to solve Interpolation

For Equal Intervals :

- Newton Forward
- Newton Backward
- central difference
 - Gauss forward
 - Gauss backward
 - Stirling central difference
 - Bessel's central difference

Unequal Intervals :

- Lagrange's interpolation
- Newton divide and difference

① Newton's Forward Difference Interpolation Formula

It is derived using a polynomial approximation.

The formula is :

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \dots$$

where, $u = \frac{x - x_0}{h},$

and h is constant difference
between successive values of x .

u is reduced variable.

Forward difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$	$\Delta^3 y_0 = \Delta^2 y_1 - \Delta^2 y_0$
x_1	y_1	$\Delta y_1 = y_2 - y_1$	$\Delta^2 y_1 = \Delta y_2 - \Delta y_1$	
x_2	y_2	$\Delta y_2 = y_3 - y_2$		
x_3	y_3			

② Newton's Backward Difference Interpolation Formula

It is derived using a polynomial approximation.

The formula is :

$$y(x) = y_n + u \nabla y_n + \frac{u(u+1)}{2!} \nabla^2 y_n + \frac{u(u+1)(u+2)}{3!} \nabla^3 y_n + \dots$$

where , $u = \frac{x - x_n}{h},$

and h is constant difference
between successive values of x .

Backward difference table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
x_0	y_0			
x_1	y_1	$\nabla y_1 = y_1 - y_0$	$\nabla^2 y_2 = \nabla y_2 - \nabla y_1$	$\nabla^3 y_3 = \nabla^2 y_3 - \nabla^2 y_2$
x_2	y_2	$\nabla y_2 = y_2 - y_1$	$\nabla^2 y_3 = \nabla y_3 - \nabla y_2$	
x_3	y_3	$\nabla y_3 = y_3 - y_2$		

③ Gauss Forward difference Interpolation Formula

The Formula is :

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{3!} \Delta^3 y_{-1} + \frac{(u+1)u(u-1)(u-2)}{4!} \Delta^4 y_{-2} \dots$$

$$\text{where, } u = \frac{x - x_0}{h}$$

and h is constant difference
between successive values of x .

$$\text{Also, } 0 < u < 1$$

Gauss Forward Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	y_{-2}	$\Delta y_{-2} = y_{-1} - y_{-2}$	$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$	$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
x_{-1}	y_{-1}	$\Delta y_{-1} = y_0 - y_{-1}$	$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$	$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
x_1	y_1	$\Delta y_1 = y_2 - y_1$			
x_2	y_2				

④ Gauss Backward Central difference Interpolation Formula

The Formula is :

$$y(x) = y_0 + u \nabla y_{-1} + \frac{u(u+1)}{2!} \nabla^2 y_{-1} + \frac{(u-1)u(u+1)}{3!} \nabla^3 y_{-2} + \frac{(u-1)u(u+1)(u+2)}{4!} \nabla^4 y_{-2} + \dots$$

where, $u = \frac{x - x_0}{h}$

and h is constant difference
between successive values of x .

Also,

$$-1 < u < 0 \Rightarrow \text{for best approximate value.}$$

Gauss Backward Table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
x_{-2}	y_{-2}				
x_{-1}	y_{-1}	$\nabla y_{-2} = y_{-1} - y_{-2}$	$\nabla^2 y_{-2} = \nabla y_{-1} - \nabla y_{-2}$	$\nabla^3 y_{-2} = \nabla^2 y_{-1} - \nabla^2 y_{-2}$	$\nabla^4 y_{-2} = \nabla^3 y_{-1} - \nabla^3 y_{-2}$
x_0	y_0	$\nabla y_{-1} = y_0 - y_{-1}$	$\nabla^2 y_{-1} = \nabla y_0 - \nabla y_{-1}$	$\nabla^3 y_{-1} = \nabla^2 y_0 - \nabla^2 y_{-1}$	
x_1	y_1	$\nabla y_0 = y_1 - y_0$	$\nabla^2 y_0 = \nabla y_1 - \nabla y_0$		
x_2	y_2	$\nabla y_1 = y_2 - y_1$			

⑤ Stirling central difference formula

Taking the mean of Gauss forward and backward formula, we obtain:

$$y(x) = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2!} \Delta^2 y_{-1} + \frac{u(u^2-1)}{3!} \left(\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right) + \frac{u^2(u^2-1)}{4!} \Delta^4 y_{-2} + \frac{u(u^2-1)(u^2-2)}{5!} \left(\frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right)$$

where, $u = \frac{x - x_0}{h}$

and h is constant difference between successive values of x .

Also, $-0.25 \leq u \leq 0.25$

Stirling central difference Table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	y_{-2}	$\Delta y_{-2} = y_{-1} - y_{-2}$	$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$	$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
x_{-1}	y_{-1}	$\Delta y_{-1} = y_0 - y_{-1}$	$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$	$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	
x_0	y_0	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
x_1	y_1	$\Delta y_1 = y_2 - y_1$			
x_2	y_2				

⑥ Bessel's Interpolation Formula

The formula is :

$$y(x) = \frac{y_0 + y_1}{2} + (u - \frac{1}{2}) \Delta y_0 + \frac{u(u-1)}{2!} \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right] + \frac{(u - \frac{1}{2}) u (u-1)}{3!} \Delta^3 y_{-1} \\ + \frac{(u+1) u (u-1) (u-2)}{4!} \left[\frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} \right] + \dots$$

$$\text{where } u = \frac{x - x_0}{h}$$

$$\text{and } \frac{1}{4} \leq u \leq \frac{3}{4}$$

Bessel's central difference Table:

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_{-2}	y_{-2}				
x_{-1}	y_{-1}	$\Delta y_{-2} = y_{-1} - y_{-2}$	$\Delta^2 y_{-2} = \Delta y_{-1} - \Delta y_{-2}$	$\Delta^3 y_{-2} = \Delta^2 y_{-1} - \Delta^2 y_{-2}$	
x_0	y_0	$\Delta y_{-1} = y_0 - y_{-1}$	$\Delta^2 y_{-1} = \Delta y_0 - \Delta y_{-1}$	$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}$	$\Delta^4 y_{-2} = \Delta^3 y_{-1} - \Delta^3 y_{-2}$
x_1	y_1	$\Delta y_0 = y_1 - y_0$	$\Delta^2 y_0 = \Delta y_1 - \Delta y_0$		
x_2	y_2	$\Delta y_1 = y_2 - y_1$			

⑦ Lagrange's Interpolation Formula

Considered the data points $(x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4) with unequal interval, then by Lagrange's formula $f(x)$ or $y(x)$ is given by :

$$\begin{aligned} y(x) = & \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} y_0 \\ & + \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} y_1 \\ & + \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} y_2 \\ & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} y_3 \\ & + \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} y_4 \end{aligned}$$

⑧ Newton divided difference formula

The formula for Newton divided difference is

$$y = f(x) = y_0 + (x-x_0) \Delta y_0 + (x-x_0)(x-x_1) \Delta^2 y_0 + \\ (x-x_0)(x-x_1)(x-x_2) \Delta^3 y_0 + \\ (x-x_0)(x-x_1)(x-x_2)(x-x_3) \Delta^4 y_0 + \dots$$

Newton Divided Difference Table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
x_0	y_0	$\Delta y_0 = \frac{y_1 - y_0}{x_1 - x_0}$	$\Delta^2 y_0 = \frac{\Delta y_1 - \Delta y_0}{x_2 - x_0}$	$\Delta^3 y_0 = \frac{\Delta^2 y_1 - \Delta^2 y_0}{x_3 - x_0}$	$\Delta^4 y_0 = \frac{\Delta^3 y_1 - \Delta^3 y_0}{x_4 - x_0}$
x_1	y_1	$\Delta y_1 = \frac{y_2 - y_1}{x_2 - x_1}$	$\Delta^2 y_1 = \frac{\Delta y_2 - \Delta y_1}{x_3 - x_1}$	$\Delta^3 y_1 = \frac{\Delta^2 y_2 - \Delta^2 y_1}{x_4 - x_1}$	
x_2	y_2	$\Delta y_2 = \frac{y_3 - y_2}{x_3 - x_2}$	$\Delta^2 y_2 = \frac{\Delta y_3 - \Delta y_2}{x_4 - x_2}$		
x_3	y_3	$\Delta y_3 = \frac{y_4 - y_3}{x_4 - x_3}$			
x_4	y_4				

Lagrange's Inverse Interpolation formula

Lagrange Inverse Interpolation is used when the value of the function y is known and the corresponding value of x is required.

$$x = \frac{(y-y_1)(y-y_2)(y-y_3)}{(y_0-y_1)(y_0-y_2)(y_0-y_3)} x_0 + \frac{(y-y_0)(y-y_2)(y-y_3)}{(y_1-y_0)(y_1-y_2)(y_1-y_3)} x_1 \\ + \frac{(y-y_0)(y-y_1)(y-y_3)}{(y_2-y_0)(y_2-y_1)(y_2-y_3)} x_2 + \frac{(y-y_0)(y-y_1)(y-y_2)}{(y_3-y_0)(y_3-y_1)(y_3-y_2)} x_3$$

Key points

① If data comes from a polynomial of degree :

- degree 1 $\Rightarrow \Delta^2 = 0$
- degree 2 $\Rightarrow \Delta^3 = 0$
- degree 3 $\Rightarrow \Delta^4 = 0$

②

Interpolation Existence & uniqueness theorem,

If n distinct data points (x_i, y_i) are given,
then :

- There exists a unique polynomial $P(x)$
- of degree at most $n-1$
- that passes through all the points.

→ At most means $\leq$, not exactly.

- ③ With n data points, we can compute differences upto order $n-1$
- ④ If a function is a polynomial of degree n , then $\Delta^{n+1} = 0$
- ⑤ When we say, $\Delta^k = 0 \rightarrow$ It means the previous difference Δ^{k-1} is constant.
- ⑥ **Missing Terms Problem :**
If number of missing value = k
then choose a degree that gives at least k independent equations.

Connect with me



[Start Practicing](#)



[i_am_arfin](#)



[Arfin Parween](#)



[Arfin Parween \(.Subscribe here too\)](#)



[Arfin Parween](#)



[Arfin Parween](#)

YouTube @StartPracticing